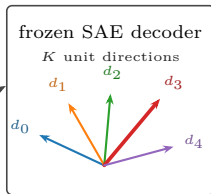
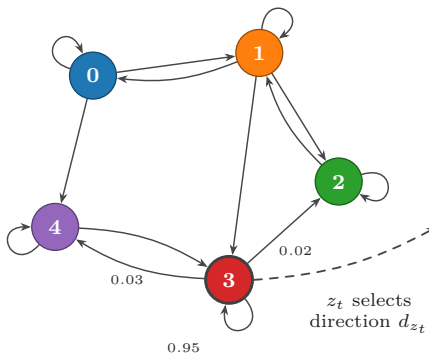


(a) plant a latent variable

any Markov chain over K states (ours: a ring, $K=8$)



z_t selects
direction d_{z_t}

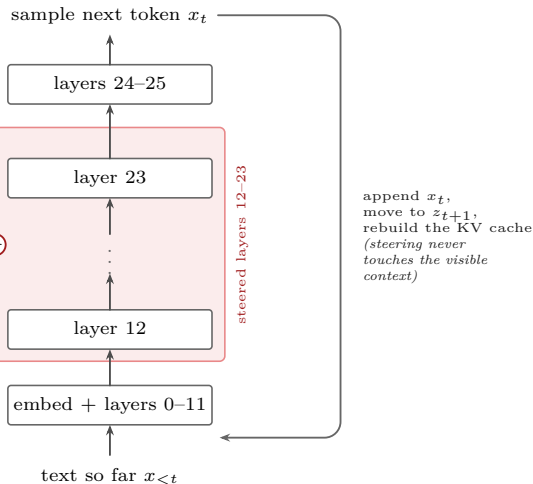
$s a_{z_t} \frac{\rho_\ell}{\rho_{12}} d_{z_t}$
added at every
layer in the band

the whole path $z_{1:T}$ is sampled before any text:



(b) steer a teacher LLM while it writes

the current state picks the direction added to the residual stream



(c) natural-looking text, secretly labeled

a real corpus excerpt — each token is colored by the state z_t that was active while it was written:

... **3** In survival, “Survival” will be presented in light of **4** the same engines of the original game. As L **3** urea writes, the game “will keep a relaxing feeling with a definite **2** imperative aim”. ...